
Financial Risk Intelligence Platform

AI/NLP risk signals from news and social media, driving tactical index rebalancing and regulator-grade portfolio stress testing

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single Docker command · zero API keys · every number measured on real data

PROBLEM & APPROACH

Risk surfaces as text first. Humans read it. Machines should too.

THE PROBLEM

- Market-moving information appears first as unstructured text: a downgrade headline, a sanctions story, an angry tweet.
- Analysts read it and decide what it means - slowly, subjectively, and only during market hours.
- The case study: build an AI/NLP engine that turns news and social text into machine-readable risk signals (sentiment, event type, impact), then prove it by acting on them.

OUR APPROACH

- Specialist models where they win: FinancialBERT for calibrated sentiment, a small LLM for per-entity understanding.
- Measure everything: 15 years of real news scored against real market moves.

WHAT THE ENGINE OUTPUTS PER HEADLINE

● Sentiment **-1 to +1**

FinancialBERT, 96.7% held-out accuracy

● Event class **6 labels**

LLM-led hybrid; survives informal text

● Impact **1 to 10**

calibrated on 19,872 real events

```
"russia nuked ukraine" -> { sentiment: -0.49, event: Geopolitical, impact: 8.0 }
```

live analyzer, adversarially tested

One engine, three layers, two downstream decisions

INGESTION

GDELT live news (15 min, keyless) · 91k FNSPID headlines (2009-23) · 10.5k real tweets · synthetic fallback

AI/NLP RISK ENGINE

entity matcher (aliases + cashtags) -> FinancialBERT sentiment [-1,1] -> per-entity LLM layer (Qwen3-4B JSON) -> event classifier -> impact = severity x conviction x corroboration

EVENTS & DECISIONS

24h event clusters with corroboration -> Module A: vol-adjusted index tilt · Module B: scenario stress + CET1 flow

SERVED AS ONE PROCESS

FastAPI (14 endpoints, SQLite WAL) serves both the REST API and the React dashboard on one port. 47 tests green · CI on every push · resumable data pipelines

Choices we can defend, and the evidence behind each

FinancialBERT for bulk sentiment

96.7%

held-out accuracy

Bake-off of 6 models on an identical held-out split: 96.7% vs 89.4% for base FinBERT. 110M params, 30ms on CPU - runs on the jury's laptop.

LLM for per-entity reading

72.4%

vs 59.8% shared-score

One headline can be positive for Apple, negative for Amazon. Qwen3-4B outputs per-entity JSON; human ceiling on this task is ~69%.

Impact calibrated on real data

19,872

events calibrated

The data overruled our priors: firm-specific news (credit, launches) moves names more than generic geopolitics. Formula stays explainable.

Regulator-grade stress testing

2,081

bps COVID depletion

DFAST/EBA scenarios + 5 historical replays (COVID, 2022, SVB), S&P-style rating migration, Vasicek PD, CET1 depletion and reverse stress.

KEY RESULTS

Every number measured on real data - 15 years of news vs real market moves

Sentiment accuracy (held-out labeled split)	96.7%	bake-off of 6 models; base FinBERT 89.4%
Directional hit rate on strong signals	56.2%	strong-signal events; literature bar 53-55%
Credit-event market reaction	+24.9 bps	t = 2.96, significant at 1%
Impact score vs realized moves	IC +0.098	calibrated severity + VIX regime; 19,872 events
Per-entity sentiment vs shared score	72.4% vs 59.8%	SEntFiN human-labeled benchmark
Rebalancer vs cap-weight (2022 bear, net of costs)	+0.9 pts	look-ahead-free walk-forward, 352 days
Vol-targeted variant (20% vol target)	-10.4%	vs -22.7% benchmark; realized vol 20.7%

WHY THIS EVIDENCE MATTERS

- No synthetic data in this table: 78k real articles + 10.5k real tweets with real timestamps, scored against real prices.
- Hit rate at/above the 53-55% bar the literature considers meaningful for daily news signals.
- We report the losing configurations too (turnover regime study) - measured, not marketed.
- Loan-book credit parameters calibrated from 1.35M real Lending Club loans.

The same workflow pattern risk teams already trust - compressed from weeks to seconds

SIGNAL

News-to-signal automation mirrors what S&P Global Market Intelligence and RavenPack sell: entity-level, calibrated, with novelty and corroboration. Ours is open and explainable.

SCENARIO

Stress testing follows the supervisory playbook (CCAR/DFAST, EBA/ICAAP): named scenarios, rating migration, Vasicek PD, CET1 depletion - the vocabulary of ICAAP submissions.

CAPITAL

The headline output is a CET1 depletion number in basis points with reverse stress: what would it take to breach the buffer. That is board-level language.

What we would build with more time - and what we would measure first

HONEST LIMITATIONS

- Shock calibration is illustrative, not supervisory-grade in magnitude fitting.
- Event clustering is rule-based (entity + type + 24h); embedding similarity would deduplicate better.
- Impact IC of 0.065 is modest by design: daily-horizon news signals live in the 0.5-2% R2 zone.
- The rebalancer trailed equal-weight in 2022; it beats its own benchmark, not every factor.

NEXT STEPS

- Intraday event windows with per-minute price data for tighter impact measurement.
- Fine-tune the event classifier on a labeled corpus (accuracy currently unmeasured - the one honest gap).
- Embedding-based deduplication and novelty scoring for event clustering.
- Streaming architecture (Kafka + point-in-time store) behind the same repository interfaces.
- Extend the universe beyond 14 names and add Indian markets (Crisil coverage).